Power Supply Induced Jitter: Introduction and Recent Advances

Chulsoon Hwang, *Senior Member, IEEE*, Jai Narayan Tripathi, *Senior Member, IEEE*, and Dan Oh

(Invited Paper)

***Abstract*—Jitter is a critical concern in modern electronic systems, particularly as semiconductor devices increasingly operate with low supply rails to minimize power consumption. This article provides a comprehensive review of power supply induced jitter (PSIJ), which refers to timing variations directly caused by fluctuations in the power supply voltage. It examines the fundamental mechanisms of PSIJ and presents generalized models for predicting jitter. System-level jitter tracking characteristics are analyzed for serial links, memory interfaces, and synchronous logic to explain the resulting jitter spectrum. The article also summarizes various presilicon PSIJ modeling techniques, including time-domain, frequency-domain, slope-based, and delay-based methods. Finally, recent advancements in PSIJ research are discussed.**

***Index Terms*—High-speed digital system, jitter, power distribution network (PDN), power supply, power supply induced jitter (PSIJ).**

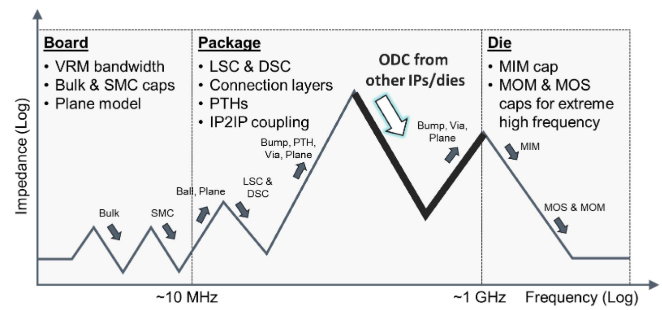


Fig. 1. Typical power distribution network impedance profile as seen from driver or receiver circuit. ODCs from other IPs or dies significantly alter the impedance profile.

I. Introduction

THIS review paper is the first one of a series aimed at summarizing the current state of the art on a specific topic of interest for the signal and power integrity community. The objective of such review papers is to consolidate the knowledge on a scientific subject, to offer a comprehensive summary on various technical aspects and applications, and to analyze which the future research path could potentially be. Thus they set the base for expert engineers dealing with design challenges, and they may serve as an education tool for those approaching such specific problems for the first time.

Research fellows from academia and industry with a long-term experience on Power Supply Induced Jitter (PSIJ) have been invited to contribute to this first paper. Prof. Chulsoon Hwang from the Missouri University of Science and Technology (USA), Prof. Jai N. Tripathi from the Indian Institute of Technology Jodhpur (India), and Dr. Dan Oh from Celestial AI (USA) discussed the physical mechanisms giving raise to the PSIJ and their modeling methodologies, with any eye on how the PSIJ is handled at a system-level design stage, and how it can be estimated at a pre-silicon phase. The recent advances concerning the modeling of nonlinear jitter sensitivity, the opportunity to include the PSIJ modeling within the IBIS Standard, and the importance of the Power Distribution Network design while taking into account its impact on driver output jitter are tackled, thus offering an overview of what could be expected from future research in this field.

Jitter is a critical concern in modern electronic system, directly impacting the accuracy of data transfer and eventually ensuring the reliable operation of high-speed digital systems. To minimize power consumption, modern semiconductor devices operate with low supply rails. This causes transistors to function outside the saturation region and deviate from an ideal current source, consequently making them susceptible to supply noise, which leads to significant signal jitter. A fundamental challenge is that supply noise does not scale well with increasing I/O speed; it is primarily dominated by the power distribution network (PDN) resonance frequency, which remains relatively constant, typically between 10 MHz and 500 MHz. As a direct result, jitter induced by fluctuations in the power supply voltage, termed power supply induced jitter (PSIJ), has become a major source of timing errors in modern high-speed I/O designs [1].

Conventional power integrity analysis often utilizes the concept of target impedance, defined based on allowable noise levels for various chip activities [2]. While design optimization frequently occurs in the frequency domain using various decoupling solutions [e.g., board-level bulk, ceramic, package, or on-die capacitors (ODCs)] as shown in Fig. 1, this approach

Received 28 August 2025; revised 2 December 2025; accepted 12 February 2026. Date of publication 16 February 2026; date of current version 4 March 2026. *(Corresponding author: Chulsoon Hwang.)*

Chulsoon Hwang is with the Missouri University of Science and Technology, Rolla, MO 65409 USA (e-mail: hwangc@mst.edu).

Jai Narayan Tripathi is with the Indian Institute of Technology Jodhpur, Jheepasani 342030, India (e-mail: jai@iitj.ac.in).

Dan Oh is with Celestial AI, Santa Clara, CA 95054 USA (e-mail: doh@celestial.ai).

Digital Object Identifier 10.1109/TSIPI.2026.3665467

can be misleading and lead to significant overdesign. This is because low- and medium-frequency noise components, which are caused by PDN peaks, are often less detrimental than high-frequency jitter occurring near the Nyquist frequency, as they can be effectively canceled by the clock signal. Therefore, focusing solely on the amplitude of noise can be misleading, and for accurate modeling of supply noise and PDN design, PSIJ must be considered [3]. Ensuring robust power integrity is intrinsically linked to controlling jitter in high-speed systems.

With increased data rates, timing budgets are becoming more stringent [4]. A crucial aspect of managing timing in high-speed digital systems is the jitter budget, which is defined as the allocation of total allowable jitter across all components of the system, including the transmitter (Tx), the interconnect or channel, and the receiver (Rx). This budgeting process ensures that the combined jitter contributions do not exceed the total available timing margins required to maintain a considerable bit error rate (BER), typically 10^{-12} or lower in high-performance systems [5], [6], [7]. Jitter budgeting plays a viral role in system design, validation, and compliance testing, and is a critical metric in standards such as peripheral component interconnect express (PCIe), universal serial bus, serial advanced technology attachment, high-definition multimedia interface, and various serializer/deserializer (SerDes) implementations [6], [7], [8], [9], [10]. A well-allocated jitter budget helps designers to optimize the performance of the system, maintain signal integrity within the system, ensure no signal component in the system consumes a disproportionate jitter margin, meet BER requirements, and clearly define the trade-offs between the performance of the Tx, channel, and Rx.

The study of PSIJ has evolved considerably since early investigations into its impact on Gigabit I/O interfaces around 2007 [11]. Concurrently, efforts to address PSIJ using standardized models also began around 2007 [12]. Over the years, research has progressed from fundamental analytical modeling of PSIJ in basic circuit elements such as CMOS inverters and buffer chains to more sophisticated system-level considerations. Initially, PSIJ analysis focused on critical timing blocks in high-speed I/O circuits, including phase-locked loops (PLLs), delay-locked loops (DLLs), and clock paths [11]. The analysis has since been extended to system-level link analysis for various I/O interfaces, such as serial links, parallel buses, and memory channels [13], [14], [15], [16]. More recently, the impact of supply noise on core logic timing has been considered [3], [17], [18], [19], [20], [21], [22]. These advancements highlight the growing recognition of PSIJ as a pervasive issue across diverse circuit and system architectures.

This article provides a comprehensive review of PSIJ, its underlying mechanisms, modeling methodologies, and practical implications in modern system design. The rest of this article is organized as follows. Section II provides a comprehensive review of PSIJ modeling including system-level jitter modeling. Section III discusses diverse techniques for estimating PSIJ in the presilicon phase of the design cycle. Section IV highlights cutting-edge developments in PSIJ research. Finally, Section V concludes this article.

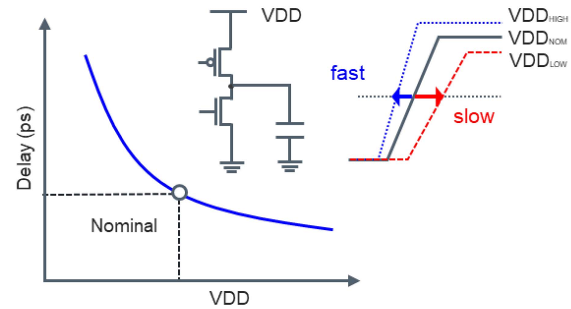


Fig. 2. Illustration of CMOS output delay versus supply voltage. Delay variation is large at low supply voltage levels.

II. PSIJ Modeling Formulation

This section provides a comprehensive review of PSIJ modeling, including a detailed formulation of system-level jitter modeling. The fundamentals of jitter sensitivity modeling are reviewed first, followed by a general formulation for system-level jitter modeling. Specific formulations for various I/O interfaces, including on-chip timing analysis, are then presented to illustrate the unique jitter-related characteristics of these interfaces.

A. *Modeling of Noise-to-Jitter Sensitivity*

Accurate modeling and analysis of PSIJ are challenging due to the inherently nonlinear nature of noise-to-jitter sensitivity. As illustrated in Fig. 2, the CMOS buffer is more delay-sensitive at low voltage. Full transistor models can be used to simulate jitter; however, they have several critical drawbacks. First, full transistor analysis requires extensive simulation time due to long data patterns and large circuit sizes. A long data pattern is necessary because power noise contains low and medium frequency components. For instance, a 56 Gbps I/O with 50 MHz power noise would require thousands of bits to simulate. Accurate delay estimation also requires *RC* extracted models of parasitic components, such as interconnects and vias, leading to large circuit sizes to simulate.

Second, analysis can only be performed at the later verification stage of design, which does not allow for early codesign opportunities between PDN and circuit designs. Early codesign analysis can allocate or trade-off between ODCs, package decoupling solutions, and circuit architectures, such as transistor types and on-chip regulation. Therefore, early PSIJ analysis based on the linear assumption is still quite useful for optimizing the overall circuit design, especially for multichip designs where various IPs are designed and integrated simultaneously. The remaining part of this section presents the simulation framework under the linear assumption. Nonlinear modeling of PSIJ will be discussed in Section IV.

Estimating jitter accumulation over data or clock paths is critical for both circuit and PDN designs. Fig. 3 illustrates the relationship between noise frequency and buffer output delay. Assuming the jitter sensitivity is linear, the supply noise induced jitter $j(t)$ can be expressed as follows:

$$j\left(t\right) = \int_{0}^{T_d} h\left(\tau\right) v_{\text{noise}}\left(t-\tau\right) d\tau \quad (1)$$

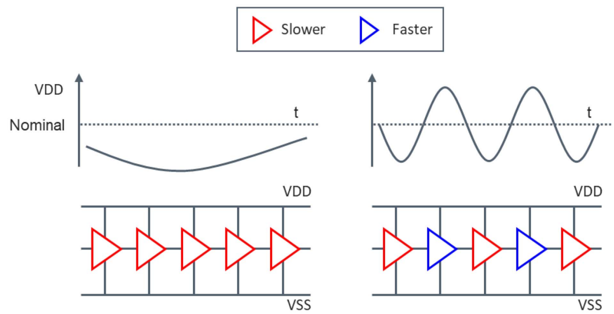


Fig. 3. Buffer delays for the low-frequency and high-frequency cases. The delay for the high-frequency case is less due to the mixture of fast and slow buffers.

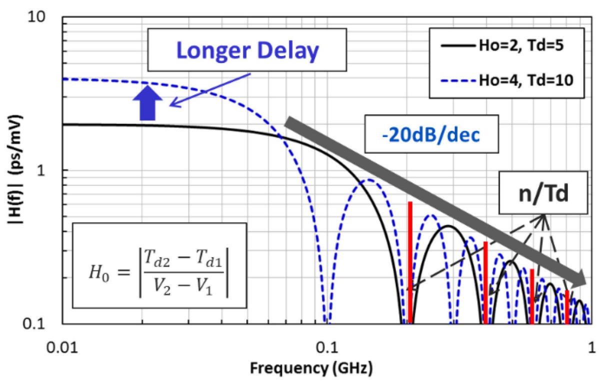


Fig. 4. Jitter transfer function of uniform linear buffer delay [17].

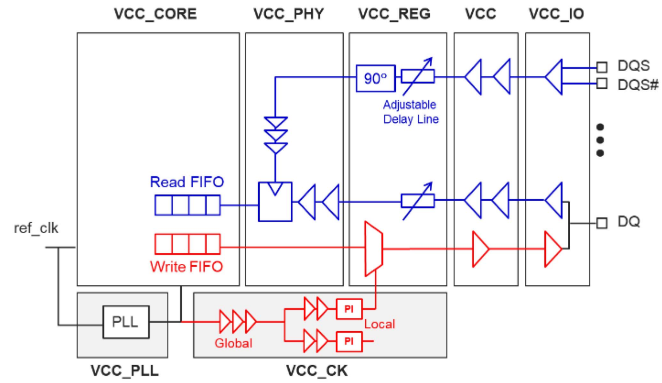


Fig. 5. Typical DDR PHY schematic for data and clock paths [16].

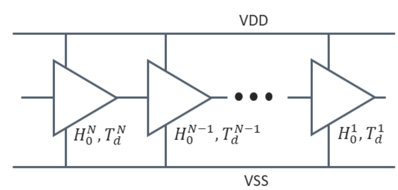


Fig. 6. Nonuniform buffer delays.

where T_d is the buffer delay. If noise-to-jitter sensitivity $h(t)$ is constant, denoted as H_o/T_d, the jitter transfer function $H(\omega)$ simplifies to a rectangular filter as follows:

$$H(\omega) = \frac{jH_0}{\omega T_d}\left[1 - e^{-j\omega T_d}\right]. \tag{2}$$

The DC sensitivity H_0 can be calculated using two static delays at two voltage levels. Fig. 4 shows the magnitude of the jitter transfer function, which is represented by the sinc function [17]. The jitter appears to be dominant at low frequency due to low PDN resonance frequencies and the sinc function nature of our jitter sensitivity. However, low-frequency jitter is often filtered out or canceled by the clock signal, making the high-frequency jitter critical in high-speed designs, as demonstrated in the following Section II-C.

B. General Formulation for Nonuniform Delay Components With Multisupply Domains

A typical data path contains various transistor types with different delays, as illustrated for a DDR PHY schematic design shown in Fig. 5 [16]. In this section, we derive the closed-form expression for the jitter sensitivity function for different delay elements in the same power domain and extend it to multiple power domains [18].

Fig. 6 shows a generic representation of N-element delays. The output jitter is the sum of the individual jitters delayed by the distance accordingly, as follows:

$$H(\omega) = H^1(\omega) + \sum_{n=2}^{N} H^n(\omega)\, e^{-j\omega\tau_0^{n-1}} \tag{3}$$

where $H^n(\omega)$ is given by (2). Now, extending to multiple power domains is straightforward by applying (3) with the inclusion of extra delay to the output stage as follows:

$$H(\omega) = \sum_{n=1}^{D} H^{\mathrm{Domain}\_n}(\omega)\, e^{-j\omega\tau_0^{n \text{ to Output}}}. \tag{4}$$

Equations (3) and (4) assume there is no jitter amplification for the through component of jitter from input to output. The impact of jitter amplification is beyond the scope of this article, and the basics of jitter amplification can be found in [15].

C. Modeling of Jitter Tracking Between Data and Clock Paths

Reducing the impact of supply noise is becoming increasingly challenging as power density on dies continues to rise due to process improvement. Fortunately, the dominant power noise components lie in the medium frequency range (10 MHz to 500 MHz). The jitter induced by this medium frequency can be mitigated by making jitter common for data and clock paths. This section describes the jitter tracking mechanism for commonly used clocking architectures such as clock forwarding I/O used in memory and parallel buses, clock-and-data-recovery (CDR)-based serial links used in PCIe, and common clock timing used in on-chip digital paths [15].

CDR can effectively remove jitter in the input signal by detecting edge variations in real time. To reduce CDR dithering, edge variations are accumulated to make phase changes, affecting CDR bandwidth. The typical bandwidth of CDR is

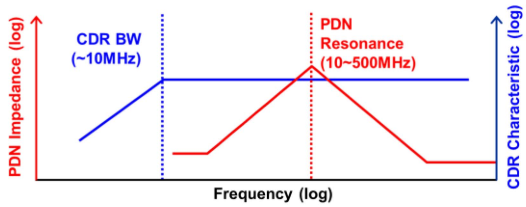


Fig. 7. Typical PDN impedance profile versus CDR loop bandwidth [1].

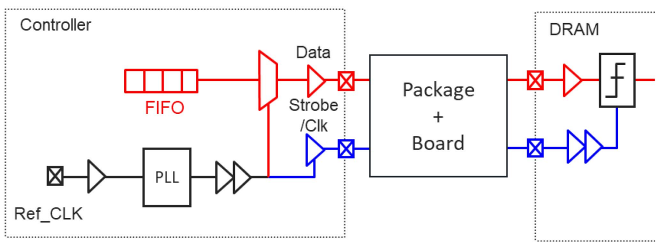


Fig. 8. Schematic representation of data and clock paths of memory interface for write operation.

below 10 MHz, which does not help reduce the medium PSIJ components. Fig. 7 shows the CDR jitter transfer function and impedance profile of a typical PDN [1].

The CDR jitter transfer function is almost the inverse of the jitter tolerance curve. The net received jitter can be represented as follows:

$$J_{\text{net}}(\omega) = J_{\text{PSIJ}}(\omega) \cdot J_{\text{CDR}}(\omega). \tag{5}$$

The loop bandwidth of CDR can be extended to higher frequencies to track PSIJ in the data signal, but the antitracking effect near the cutoff frequency must be carefully analyzed.

Clock forwarding schemes are widely used in high-speed I/O interfaces, such as memory buses (DDR and HBM), parallel interfaces (Intel's QuickPath Interconnect (QPI), AMD's Infinity Fabric). The jitter due to on-chip data path is becoming more important for upcoming die-to-die interfaces, such as, universal chiplet interconnect express, bunch of wires, etc., because the off-chip channel is relatively simple and short; on the other hand, the on-chip data path is longer due to an increase in I/O counts. Fig. 8 shows the generic schematic representation of a typical clock forwarding interface, particularly for memory I/Os [15].

The jitter at the receiver can be written as follows:

$$J_{\text{Setup}}(\omega) = J_{\text{Data}}(\omega) - J_{\text{Strobe}}(\omega) \cdot e^{-j\omega T_{\text{offset}}} \tag{6}$$

where T_{offset} represents the sampling clock offset from the transmit clock for the sampled data signal, which is required as the clock path has a longer delay than the data path. Similarly, the hold side expression can be given as follows:

$$J_{\text{Hold}}(\omega) = J_{\text{Data}}(\omega) \cdot e^{-j\omega T_{\text{bit}}} - J_{\text{Strobe}}(\omega) \cdot e^{-j\omega T_{\text{offset}}}. \tag{7}$$

Here, the data signal is shifted by a bit time to sample at the hold side. The negative jitter value contributes to the margin loss for both the setup and hold cases. Fig. 9 shows the jitter cancellation due to the common jitter on data and clock signals [19]. Eye opening is significantly wider after triggering with clock (strobe)

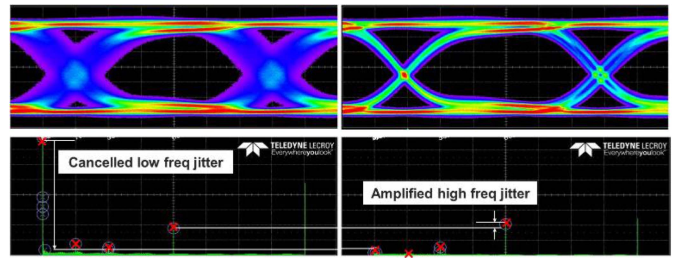


Fig. 9. DDRx eye diagrams triggered by an ideal clock and strobe signal.

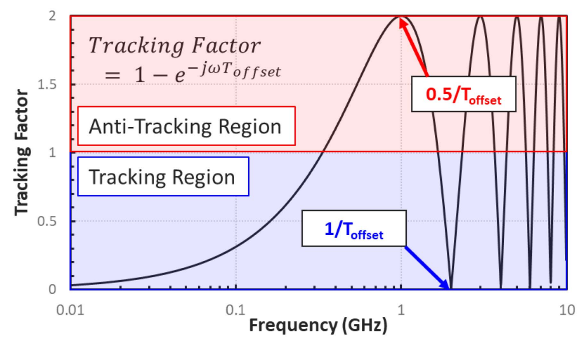


Fig. 10. Tracking filter (period jitter filter) of clock forwarding architecture ($T_{\text{offset}} = 500$ ps) [18].

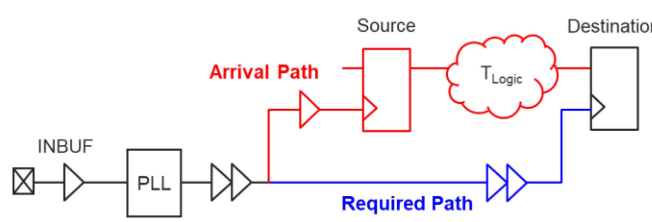


Fig. 11. Digital logic timing path example.

signal. It is important to notice that the high-frequency noise is increased due to the antitracking effect. Fig. 10 illustrates this jitter tracking behavior. Any common jitter between the data and clock paths are subject to this tracking curve [18].

Digital logic uses synchronous clocking, and timing constraints are validated with static timing analysis (STA). STA uses various delay models at different supply voltage levels to account for supply variation and noise. Most modern chips generate significantly higher supply noise than the ones accounted for in STA models [3]. To account for this large supply noise, enhanced STA analysis with PSIJ was proposed [3], [17], [18], [19]. A generic logic timing path is shown in Fig. 11 [3], and the logic timing jitter can be expressed similar to the clock forwarding case as follows:

$$H_{\text{Setup}}(\omega) = H_{\text{Arrival}}(\omega) - H_{\text{Required}}(\omega) \cdot e^{-j\omega n T_{\text{CLK}}} \tag{8a}$$

$$H_{\text{Hold}}(\omega) = [H_{\text{Arrival}}(\omega) - H_{\text{Required}}(\omega)]\, e^{-j\omega n T_{\text{CLK}}} \tag{8b}$$

where n is one for a single cycle. In the above equations, the expressions are written with the jitter sensitivity functions as the digital blocks are all powered by the same supply. If the

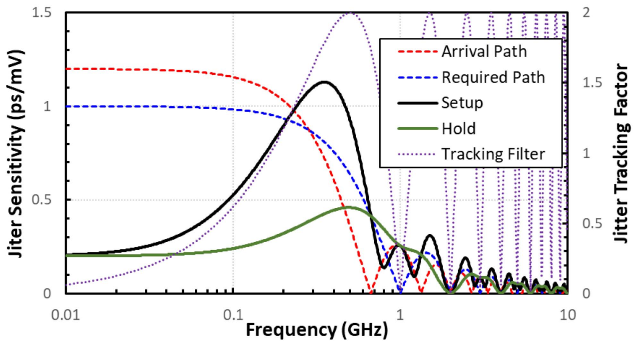


Fig. 12. Setup and hold jitter sensitivity of data path (Arrival path delay and sensitivity: 1.5 ns and 1.2 ps/mV, required path delay and sensitivity: 1 ns and 1.0 ps/mV, 1 GHz frequency, nT_{CLK} is one period.

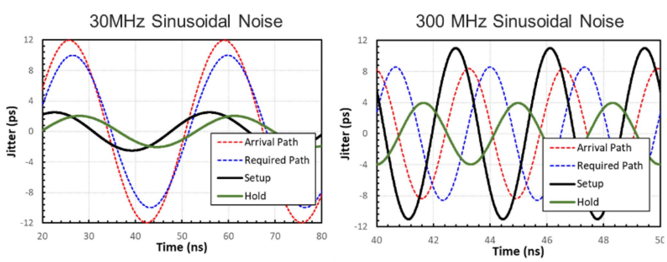


Fig. 13. 20 mVpp sinusoidal noise sources of 30 MHz and 300 MHz were applied to Fig. 11 case.

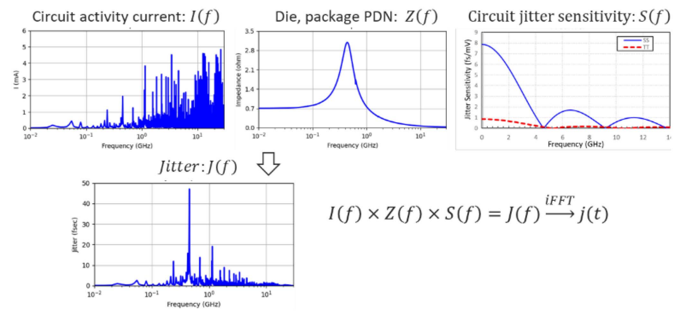


Fig. 14. 20mVpp sinusoidal noise sources of 30 MHz and 300 MHz were applied to Fig. 11 case.

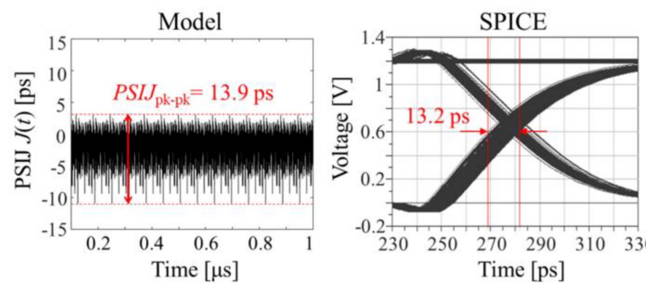


Fig. 15. PSIJ comparison results for 3-stage clock buffers and I/O driver under 128 I/O aggressor conditions [24].

arrival and required paths have the same delay, the setup side sensitivity becomes a period jitter filter shown in Fig. 10, and the hold side sensitivity is zero. In real circuits, there is always additional delay on the arrival data path due to combinatory logic blocks between flops, as shown in Fig. 11.

The sensitivity curves for the setup and hold for the data path delay are illustrated in Fig. 12. As shown in this figure, the setup side jitter sensitivity is much worse than the hold side because the jitter tracking between the arrival and required path is worse due to the additional phase difference. Fig. 13 demonstrates the time-domain example of the jitter tracking results between the arrival and required paths. 30 MHz and 300 MHz sinusoidal noise sources were injected in the previous case. For the 30 MHz low-frequency jitter on the arrival and required paths is effectively removed for both setup and hold cases. For the 300 MHz case, the hold case shows the jitter tracking, but the setup side showed antitracking, emphasizing the importance of controlling high-frequency power noise.

D. PSIJ Modeling Framework

One of the most crucial aspects of PSIJ modeling is its capability to optimize chip and package design during the early architecture stage of chip development. This is particularly vital for modern advanced package devices, where integrated components may share a common supply, making early chip/package codesign essential for the first-time success. Fig. 14 illustrates the PSIJ modeling and simulation process in the frequency domain. While this process assumes a linear jitter sensitivity function, it suffices for optimizing the overall PDN design. For more precise postlayout time-domain simulation, a full transistor model is recommended, though due to design complexity, it can only be used to verify small blocks.

III. Various PSIJ Modeling Techniques

While designing the system, it is necessary to estimate jitter in the presilicon phase of the design cycle to ensure it stays within the specified jitter budget. Full simulations using EDA tools can be computationally expensive for such tasks; therefore, efficient modeling techniques are required to estimate jitter. This section discusses diverse techniques for estimating PSIJ, including analytical, semianalytical, as well as the input/output buffer information specification (IBIS) model-based techniques. The jitter estimation techniques can also be categorized based on whether the analysis is done in the time domain or frequency domain.

A. Time-Domain Analysis

As jitter is associated with the timing inaccuracies in the transition edges, it can be estimated by the transient analysis of the system response. If there is noise in the power supply/ground, the transient response will be able to capture the variations in transition edges accordingly. There are several techniques available in the literature [24], [25]. In [24], a device parameters-based modeling of PSIJ is presented for a case study of a CMOS inverter. The PSIJ is estimated analytically based on the values of device parameters and the noise present in the power supply. For performing this, the output response of the inverter was divided into five regions based on the operating modes of the transistors. For each region, the time domain equations were derived in the presence of noise, and later all these regions were concatenated. A closed-form expression for the timing interval error (TIE) is derived, considering multitone noise sources (v_s) on top of the power supply (V_{DD}). These multitone components are derived from the Fourier series of the periodic power supply noise. The output jitter is then quantified as the peak-to-peak value of the computed TIE as follows:

$$\text{PSIJ} = \max\{\text{TIE}\} - \min\{\text{TIE}\}. \tag{9}$$

where

$$\text{TIE}_j = \tau + \left(V_{23_j} - \frac{V_{DD}}{2} + B\right) \cdot \left(\frac{1}{A}\right) - T_0 \tag{10}$$

where T_0 represents the time instant at which $V_{\text{out}} = V_{DD}/2$, as derived from (38) in [24] under ideal supply conditions ($v_s = 0$). The difference between the maximum and minimum values of TIE defines the PSIJ at the output. Here, V_{23_j} is defined as follows:

$$\begin{aligned} V_{23_j} = V_{DD} - &\sum_{k=1}^{\infty} b_{nkj} \left(\frac{(V_{DD} - |V_{\text{TP}}| - V_{\text{TN}})\,\tau}{V_{DD}}\right)^k + V_{12_j} \\ &+ \sum_{n=1}^{N} A_n \sin\phi_{nj} - \frac{C_{gd}\,(V_{DD} - |V_{\text{TP}}|)}{C_L + C_{gd}} \\ &+ \frac{\beta_n \tau (V_{DD} - |V_{\text{TP}}| - V_{\text{TN}})^3}{6\,(1+\delta_n)\,(C_L + C_{gd})\,V_{DD}} \end{aligned} \tag{11}$$

where the details of notations used in (11) can be found in [24].

In [25], a comprehensive analytical approach to model deterministic jitter at the output of CMOS inverters, considering various noise sources such as power supply noise, ground-bounce noise, and data noise, is presented. The estimation of jitter is performed by calculating the peak-to-peak deviation in the transition edge of the output of the inverter, taking into account the impact of each noise source individually as well as in combination. Next, in [26], a concept of variability-aware jitter modeling was proposed. It demonstrates that the process variability effects need to be modeled along with deterministic jitter to get an estimate of extreme conditions of deterministic jitter. These analytical techniques present a reliable and experimentally validated method for estimating deterministic jitter in CMOS circuits. The consistency in results with both simulation and measurement confirms its utility as an effective tool for timing analysis and signal integrity evaluation in high-speed VLSI design.

B. Frequency-Domain Analysis

In frequency domain analysis-based techniques, jitter is estimated first in the frequency domain and then translated into the time domain. These techniques use information about the spectrum of current in the system and the impedance of the system. Based on this information, power supply noise can be estimated, which is eventually used with a jitter transfer function to estimate jitter. One such example is presented in [27]. The work reported in [27] presents modeling, analysis, and optimization of PSIJ in an HBM I/O interface. Each I/O region's complete PSIJ spectrum is modeled at an observation port i. The sum of the jitter contributions from the clock buffers, $J_{cll,i}(f)$, and the I/O drivers, $J_{\text{driver},i}(f)$, defines the PSIJ spectrum, $J_i(f)$, at port i as follows:

$$\begin{aligned} J_i(f) &= J_{\text{clk},i}(f) + J_{\text{driver},i}(f) \\ &= S_{\text{clk},i}(f)\,V_i(f) + S_{\text{driver}}(f)\,V_i(f) \\ &= S_{\text{clk},i} + (f)\,Z_{\text{sum}i}(f)\,I(f) \\ &\quad + S_{\text{driver},i}(f) Z_{\text{sum}i}(f) I(f). \end{aligned} \tag{12}$$

Finally, PSIJ in the time domain, $J_i(t)$, is obtained by applying the inverse Fourier transform (IFFT) to the frequency domain spectrum: $J_i(t) = \text{IFFT}(J_i(f))$. The peak-to-peak PSIJ, denoted as $J_{\text{pk-pk},i}$, is then calculated as the difference between the maximum and minimum values of $J_i(t)$: $J_{\text{pk-pk},i} = \max(J_i(t)) - \min(J_i(t))$. The detailed description of this method is discussed in [24].

C. Slope-Based Techniques

The work in [28] presents a methodology to estimate PSIJ for the voltage-mode driver circuit. This approach is based on separating the large- and small-signal output responses of the circuit and then combining them to obtain the overall response of the system. Simplified expressions for PSIJ are derived based on the slope of the rising and falling edges of the output signal. The method, abbreviated as EMPSIJ, estimates jitter as follows:

$$J_r = \frac{v_{\text{diff}}(t_m)}{\alpha} \tag{13}$$

where α is the slope of the output when there is no noise present in the circuit, and t_m is the time stamp of the midpoint (V_{DD}/2) of the differential output. Thus, the generalized jitter expression considering all the noise sources can be written as follows:

$$J_r = \frac{1}{\alpha} \sum_{i=1}^{N} \sum_{j=0}^{\infty} \eta_{ij} v_{n_{ij}}(t_m) \tag{14}$$

where η_{ij} is the generalized transfer function from the *i*th noise source to the output for a particular frequency *j* and *N* is the total number of noise sources. $v_{n_{ij}}$ is the *j*th frequency component of the *i*th noise source.

Using the EMPSIJ approach, a significant reduction in computational complexity while maintaining accuracy is achieved, making it a practical solution for jitter estimation in high-speed VLSI circuits. In [29], this slope-based approach is further extended for jitter estimation in a chain of inverters. The chain of CMOS inverters is widely used in many practical applications such as delay lines, clock distribution, ring oscillators, etc. The analysis in [29] is done in two parts to apply the EMPSIJ method for the circuit of Fig. 16.

First, the TIE, denoted as $(J_r)_M$, is calculated at the *M*th stage output (where all transistors remain in the saturation region) using expressions presented in [29]. Next, for the (*M*+*1*)th stage, $(J_r)_{M+1}$ is assumed to be the sum of $(J_r)_M$ and the TIE due to noise from the power supply and ground. It is assumed that the delay δ_t from stage M to stage $M+1$ remains constant for $\beta_p = \beta_n$, where β_p and β_n are the gain factors for pMOS and

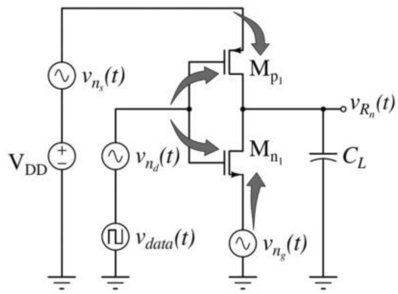


Fig. 16. Multiple noise sources and their paths of noise in an inverter.

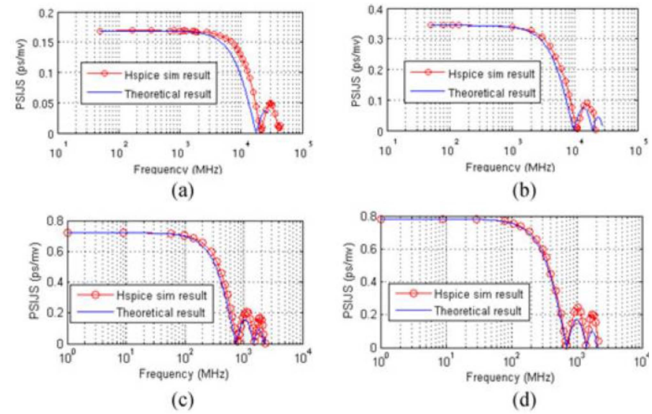


Fig. 17. Plots of PSIJ sensitivity from Hspice simulation results. (a) Buffer chain 1, (b) buffer chain 2, (c) buffer chain 3, and (d) buffer chain 4.

nMOS transistors, respectively. This delay is nearly constant across stages after the *M*th stage. The noise from the power supply and ground exhibits similar transfer functions to that of a single inverter stage. The output responses due to these noise sources are then combined to obtain the jitter at the output of the (*M+1*)th stage. Thus, the TIE at the output of the (*M+1*)th stage can be calculated as follows:

$$(J_r)_{M+1} = \frac{v'_{rn(M+1)}}{\alpha_{M+1}} + (J_r)_M. \tag{15}$$

By following similar steps as outlined above, the TIE at the output of the *i*th stage can be expressed as follows:

$$(J_r)_i = \frac{v'_{rn(i)}}{\alpha_i} + (J_r)_{i-1}, \quad i > M. \tag{16}$$

The total output jitter for *N* stages can be formulated as follows:

$$J_N = \max\{(J_r)_N\} - \min\{(J_r)_N\}. \tag{17}$$

D. Delay-Based Techniques

Due to power supply noise, variations occur in the switching transitions of circuits, which subsequently affects the delay of the circuit. Considering this phenomenon, several delay-based modeling methods have been reported in the literature. The fundamental concept behind these methods is to model the perturbations in rising and falling edges caused by supply noise with respect to the inherent delay of the circuits.

1) Jitter Sensitivity Based on Propagation Delay: A simple and accurate expression for PSIJ sensitivity in a CMOS buffer chain is presented in [30]. Notably, this expression neither requires detailed knowledge of the architecture of buffer chain nor its electrical parameters, making it suitable for rapid estimation of jitter induced by supply noise. Jitter sensitivity is defined as the ratio of the jitter generated by a single-frequency supply noise signal to the amplitude of that noise signal [30]. Accordingly, the magnitude of PSIJ sensitivity is given as follows:

$$|\text{Jitter sensitivity}(f)| = \frac{T_{P,\max}(f) - T_{P,\min}(f)}{V_n - (-V_n)} \tag{18}$$

where $\mathrm{T}_{P\max}$ and $\mathrm{T}_{P\max}$ are the maximum delay (when V_{supply} dips to minimum) and minimum delay (when V_{supply} is at nominal or high), respectively. This formula allows designers to calculate how much jitter is induced for a given supply noise amplitude, using just the DC delay swing over known voltage variation. Fig. 17 shows the plots of jitter sensitivity functions with respect to frequency for different buffer chains.

2) Delay Based Estimation of Time Interval Error: In [25], another delay-based modeling approach for estimating PSIJ is introduced. The work in [25] presents a case study of modeling PSIJ in a current-mode driver circuit. This approach is a semianalytical technique, which requires one bit simulation to accurately estimate the inherent delay of the circuit in presence of a clean power supply. Based on this delay, the output response of the driver is approximated as a first-order model involving an *RC* time constant. The expressions are further developed considering the impact of deterministic noise on the delay, and eventually the expression for TIE is derived. The noise transfer functions are used in this approach. This technique is useful as it uses a first-order approximation, which results into simple expressions that can be physically relatable. In this work, it is also demonstrated that both the slope-based approach and the delay-based approach led to the same expressions when linear approximation is considered. This establishes the validity of both approaches and builds a sound theoretical foundation to understand PSIJ.

IV. Recent Advances

A. Nonlinear Jitter Sensitivity

Conventional PSIJ modeling and estimation assume a linear time invariant (LTI) system in which induced jitter has a linear relation with power supply noise. Based on this assumption, linear jitter sensitivity can be extracted and modeled in the frequency domain and then used to estimate jitter as described in Section II. However, in real circuits, the interaction between power supply noise and the system is influenced by the nonlinearities in the circuit components. It is well known that the linear jitter sensitivity approximation is valid only when the noise amplitude is small, usually less than 5% to 10% of the nominal supply voltage [32], [33], [34], [35]. For example, in a DDR interface PSIJ case introduced in [34], jitter prediction using linear jitter sensitivity showed a 10% error for a 5%

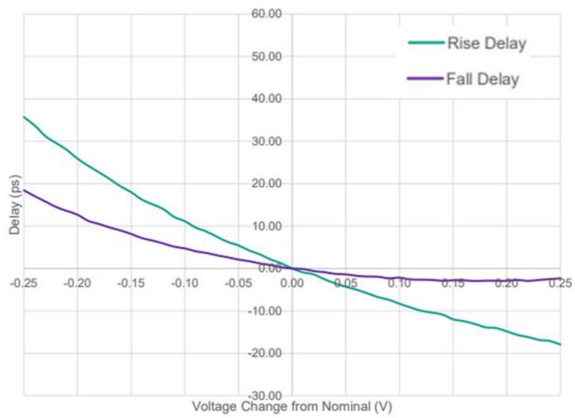


Fig. 18. Nonlinear PSIJ behavior of DDRx DQ buffer [35].

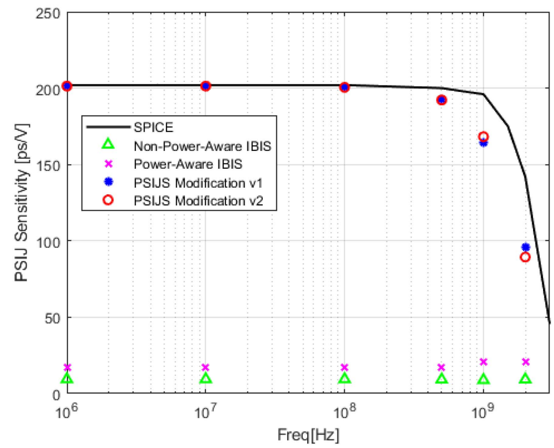


Fig. 19. PSIJ sensitivity comparison between different IBIS models from [40].

supply fluctuation, while the error surged to ∼30% for a 15% supply fluctuation. Another example is demonstrated in Fig. 18 [35], where simulated propagation delay of DDRx DQ buffer shows a nonlinear relationship to the power supply noise change. With nonlinear jitter sensitivity, the jitter is no longer linearly proportional to the amplitude of power supply noise, and the timing variance with multitone power noise can no longer be the sum of the timing variance with single-tone power noise.

Recently, a method to use lookup tables to predict PSIJ, considering the nonlinear effect, is introduced in [34]. The jitter value given by the lookup table is essentially for DC supply voltage change. As done in the linear jitter sensitivity derivation, the final jitter at the buffer output is the time-averaged jitter over the propagation delay, which can then be derived as follows [34]:

$$j\left(t\right) = \int_{0}^{T_d} D\left[V_{DD}\left(t-\tau\right)\right] d\tau \tag{19}$$

where $T_{\mathrm{pd}\,0}$ is the nominal propagation delay of the path supplied by V_{DD} and $D(V_{DD})$ is the lookup table.

For comparison, the jitter with a linear assumption is shown below:

$$j\left(t\right) = -\frac{\Delta T_{pd}/\Delta V_{DD}}{T_{pd0}} \int_{0}^{T_d} (V_{DD}(t-\tau) - V_{DD0}) d\tau \tag{20}$$

where V_{DD0} is the nominal power supply voltage and ΔT_{pd} and ΔV_{DD} are the delay change and power supply variation, respectively.

There are two differences between (19) and (20): the DC jitter sensitivity is assumed to be linear, so it is factored out from the integration, and only power noise fluctuation from its nominal voltage is considered. The lookup table implicitly includes the DC jitter sensitivity in it. Equation (20) is essential the same as (1) with different notations for noise-to-jitter sensitivity.

B. Jitter Modeling in IBIS Standard

IBIS is an industry standard method to electronically transport modeling data between semiconductor vendors, EDA tool vendors, and end users. It describes the behavior of the buffers using a black box model and thus provides more computationally efficient simulations and intellectual property protection. Since the first specification was released in the early 1990s, the IBIS model has undergone continuous improvement, and it is now widely used in the industry for signal integrity simulations.

Before the release of IBIS Version 5.0 in 2008, IBIS models focused primarily on signal integrity and did not account for power integrity. Power-aware IBIS models were first introduced with IBIS Version 5.0, which was a major milestone for the IBIS specification that allowed the modeling of power-related effects. One major update is to consider current flowing into the power/ground rails, allowing a more accurate power/ground bounce analysis and simulation [12]. Another update is to consider the "gate modulation effect" that the drive strength of the gate may vary depending on the value of supply voltage with the simultaneous switching output noise [36]. The gate modulation effect is included in the IBIS model by modifying switching coefficients depending on the power noise.

The power-aware IBIS model, however, can only consider power noise effects at the final buffer stage. To account for the PSIJ occurring in the predriver stage, an IBIS switching coefficient modification algorithm was proposed by considering the power-rail-voltage time-averaged effect [37], [38]. The proposed algorithm has been improved to encompass the nonlinear jitter sensitivity [39], [40]. IBIS simulation results using different versions of models are shown in Fig. 19. The example shown here is for a buffer with a predriver stage under power noise and with AC noise amplitude of 50 mV. In addition to the switching coefficient modification, a key word has been proposed to consider the holistic jitter impact from all power supplies, such as predrivers, clock distribution tree, and other logic blocks [41], [42], which focuses on system-level jitter simulation.

C. Jitter Aware PDN Design

As discussed earlier, optimizing the system in the frequency domain solely based on a target impedance may result in overdesign. Therefore, efforts have been made to incorporate PSIJ sensitivity into PDN performance evaluations [43], [44], [45].

In [44], the simulated PSIJ sensitivity is transformed into the time domain using the inverse Fourier transform and then convolved with power noise to efficiently compute jitter in the time domain. The calculated jitter is subsequently used as a metric for PI analysis of the SerDes interface. The method demonstrated a 360X faster simulation time compared to transistor-level simulations, which usually take several hours to days.

The calculated jitter is also utilized in the PDN optimization process. For instance, during decoupling capacitor placement optimization, PSIJ sensitivity is considered, and the calculated jitter serves as a key performance metric [44], [45]. PSIJ sensitivity is combined with PDN impedance (with a known current profile) in either the time or frequency domain. Since jitter, here meaning continuous time interval error, is typically analyzed in the frequency domain, it must be transformed into the time domain for proper evaluation. In [44], jitter-aware decoupling capacitor optimization demonstrated significantly better jitter performance than conventional target impedance-based optimization.

To establish a stronger correlation between PDN impedance parameters and PSIJ, a jitter-aware target impedance has been proposed [46], [47]. It is well known that conventional *R*- or *RL*-type target impedance approaches often lead to overdesign issues. Improved target impedance concepts have been introduced [48], [49], but these methods only account for the maximum allowable power supply voltage fluctuations. In [46] and [47], PSIJ sensitivity is incorporated into the target impedance derivation, making it more informative and enabling a more direct correlation between PSIJ and PDN target impedance. The PSIJ of the output buffer is analytically derived and combined with PDN parameters (assuming a known equivalent PDN model). As a result, a PDN designed to meet the target impedance ensures compliance with the jitter specification.

D. Artificial Neural Network Applications in PSIJ

As machine learning/artificial intelligence is revolutionizing industries and reshaping almost every technology, its applications in PSIJ are not an exception. The utilization of artificial neural networks (ANNs) for PSIJ has been focused on replacing the transistor-level SPICE simulation with neural networks to shorten the simulation time while achieving reasonable accuracy.

An early attempt to use ANNs to model the power supply noise effect on the buffer output was made about two decades ago [50], [51]. Mutnury et al. [50], [51] extended recurrent neural network (RNN) models to multiple ports to capture power supply noise when multiple drivers are switching simultaneously. RNN models are suitable for highly nonlinear patterns with memory or feedback, making it a good choice for complex drivers.

In [52], a multilayer perceptron (MLP) neural network is used to model PSIJ of different DC bias. The trained neural network becomes fast and efficient in representing the DDR IO buffer during power integrity simulation and design. The algorithm was implemented in Verilog-A and showed about 120× faster speed than the transistor-level simulations while showing an error less than 1%.

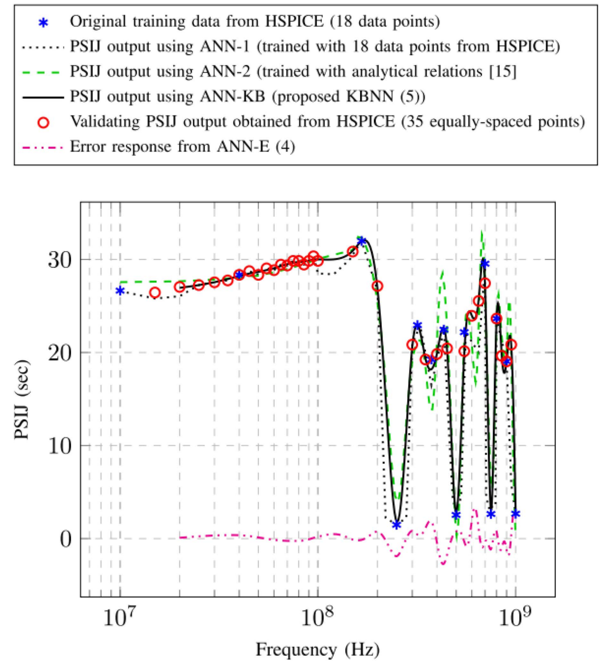


Fig. 20. Comparison of PSIJ response using the ANNs versus the conventional approach (HSPICE) [53].

ANNs require a large amount of training data that can be extracted from SPICE simulations or surrogate methods. To reduce training time costs, knowledge-based networks were used to estimate PSIJ performance in [53] and [54]. A large amount of training data is collected using analytical models and a small amount of accurate data (ground truth) using a circuit simulator. It demonstrated considerable speedup for comparable accuracy. A comparison of PSIJ response using the trained ANN versus the conventional approach (HSPICE) is shown in Fig. 20.

E. PSIJ in 2.5-D and 3-D ICs

Driven largely by the rise of artificial intelligence, chip-package systems face an ever-increasing demand for high bandwidth, driving the adoption of 2.5-D and 3-D IC integration. When multiple dies share the same PDN, transient current can increase sharply, generating significant supply noise. Although the short interconnects in these architectures help mitigate traditional signal-integrity challenges, PSIJ has emerged as one of the dominant performance bottlenecks [22], [55], [56].

Recent research highlights the need for accurate system-level PSIJ modeling, particularly for the parallel interfaces widely used in 2.5-D and 3-D ICs. Because chip behavior, package SI/PI characteristics, and interface circuits are tightly coupled, analysis methods must consider them jointly rather than as an isolated domain. Efficient empirical and analytical models are also essential to enable early-stage PSIJ prediction without relying solely on slow transistor-level simulations.

In [22], supply noise and timing jitter in a heterogeneous integrated system using Intel's embedded multidie interconnect bridge are modeled through efficient empirical analysis and characterization techniques, with particular focus on PSIJ. In addition to conventional PSIJ modeling methodologies, the work investigates source-synchronous clocking, a common feature of such parallel interfaces. In [55], PSIJ for a 4 Gbps high bandwidth memory (HBM) I/O interface is examined, identifying PSIJ as a key noise component in highly integrated chip-package systems and parallel interfaces. To achieve accurate prediction and effective optimization in the predesign stage, the PDN, channel, and Tx/Rx clocking circuits are integrated into a unified model. It has been noted that the critical frequency range driving PSIJ lies between 800 MHz to 5 GHz, since lower frequency jitter is largely tracked and canceled by the inherent source-synchronous nature of the parallel interfaces. Increasing on-chip decoupling capacitance is highlighted as one of the most effective mitigation strategies for system-level PSIJ reduction. PSIJ in chiplet-based 2.5-D integrated systems is also discussed in [56], where detailed modeling of the system transfer impedance, simultaneous switching current, jitter sensitivity function, and channel amplification factor enables accurate assembly and validation of the final PSIJ profile.

V. Conclusion

Jitter remains a paramount concern in modern electronic systems, directly affecting data transfer accuracy and the reliable operation of high-speed digital systems, and PSIJ becomes a major source of signal jitter. Conventional power integrity analysis, often based on target impedance, while useful, can lead to significant overdesign. Therefore, focusing solely on noise amplitude is misleading; accurate modeling of supply noise and PDN design necessitates the explicit consideration of PSIJ. This article has provided a comprehensive review of PSIJ. Detailed formulations for PSIJ modeling including system-level architectures, jitter budgeting for components, various PSIJ modeling techniques, and recent advances in PSIJ research were discussed. Robust PSIJ modeling and analysis are indispensable for optimizing chip and package design, ensuring fast and reliable operation of high-speed digital systems.

References

[1] Y. Shim and D. Oh, "Practical issues and solutions for supply induced jitter decomposition," in *Proc. DesignCon*, 2016.

[2] H. Barnes, "Power integrity fundamentals: Impedance vs. frequency," *Signal Integrity J.*, May 6, 2021. [Online]. Available: https://www.signalintegrityjournal.com/blogs/12-fundamentals/post/2108-power-integrity-fundamentals-impedance-vs-frequency

[3] D. Oh, A. Razmadze, and K. Chandrasekar, "Power integrity analysis for core logic blocks," in *Proc. IEEE 22nd Conf. Elect. Perform. Electron. Packag. Syst.*, 2013, pp. 79–82.

[4] J. N. Tripathi, V. K. Sharma, and H. Shrimali, "A review on power supply induced jitter," *IEEE Trans. Compon. Packag. Manuf. Technol.*, vol. 9, no. 3, pp. 511–524, Mar. 2019.

[5] D. Oh et al., "Prediction of system performance based on component jitter and noise budgets," in *Proc. IEEE Elect. Perform. Electron. Packag.*, 2007, pp. 33–36.

[6] PCI Express Card Base Specification, Rev. 2.0, PCI-SIG, Dec. 20, 2006. [Online]. Available: https://pcisig.com

[7] PCI Express Card Electromechanical Specification, Rev. 2.0, PCI-SIG, Apr. 11, 2007. [Online]. Available: https://pcisig.com

[8] "USB 3.0 jitter budgeting white paper, revision 0.5, USB implementers forum," 2009. [Online]. Available: https://www.usb.org/document-library/usb-30-jitter-budgeting

[9] "USB 3.2 revision 1.1, USB implementers forum," Jun. 2022. [Online]. Available: https://www.usb.org/document-library/usb-32-revision-11-june-2022

[10] G. Edlund, "PCI express timing margins," in *Proc. DesignCon*, 2008, pp. 1–24.

[11] R. Schmitt, H. Lan, C. Madden, and C. Yuan, "Investigating the impact of supply noise on the jitter in gigabit I/O interfaces," in *Proc. IEEE Elect. Perform. Electron. Packag.*, 2007, pp. 189–192.

[12] A. Muranyi, A. Girardi, G. Bernardi, R. Izzi, and B. Ross, "BIRD 98.3: Gate modulation effect," Jul. 2007. [Online]. Available: https://ibis.org/birds/bird98.3.txt

[13] D. Oh and S. Chang, "Clock jitter modeling in statistical link simulation," in *Proc. 18th Conf. Elect. Perform. Electron. Packag. Syst.*, 2009, pp. 49–52.

[14] D. Oh, S. Chang, and J. Ren, "Hybrid statistical link simulation technique," *IEEE Trans. Adv. Packag.*, vol. 1, no. 5, pp. 772–783, May 2011.

[15] D. Oh and C. Yuan, *High-Speed Signaling: Jitter Modeling, Analysis, and Budgeting*. Englewood Cliffs, NJ, USA: Prentice Hall, 2012.

[16] Y. Shim and D. Oh, "Optimizing DDR4 PHY design to reduce the system-level jitter impact by supply noise," in *Proc. DesignCon*, 2014.

[17] D. Oh and Y. Shim, "Power integrity analysis for core timing models," in *Proc. IEEE Symp. Electromagn. Compat.*, 2014, pp. 833–838.

[18] D. Oh and Y. Shim, "A novel approach to model dynamic noise in static timing analysis," in *Proc. DesignCon*, 2015.

[19] Y. Shim and D. Oh, "Impacts of dynamic noise in multi-core or SoC designs," in *Proc. DesignCon*, 2016.

[20] Y. Shim and D. Oh, "System level modeling of timing margin loss due to dynamic supply noise for high-speed clock forwarding interface," *IEEE Trans. Electromagn. Compat.*, vol. 58, no. 4, pp. 1349–1358, Aug. 2016.

[21] H.-S. Kang, G. Chen, A. Hashemi, W. S. Choo, D. Greenhill, and W. Beyene, "Simulation and measurement correlation of power supply noise induced jitter for core and digital IP blocks," in *Proc. DesignCon*, 2019.

[22] W. T. Beyene, H.-S. Kang, A. Hashemi, G. Chen, and X. Liu, "Noise and jitter characterization of high-speed interfaces in heterogeneous integrated systems," *IEEE Trans. Compon. Packag. Manuf. Technol.*, vol. 11, no. 1, pp. 109–117, Jan. 2021.

[23] S. Park, H. An, S. Jeon, G. Kim, and D. Oh, "Chip/package co-design analysis of advanced D2D interface using a statistical link simulator," in *Proc. Electron. Compon. Technol. Conf.*, 2021, pp. 1844–1848.

[24] P. Arora, J. N. Tripathi, and H. Shrimali, "Device parameter-based analytical modeling of power supply induced jitter in CMOS inverters," *IEEE Trans. Electron Devices*, vol. 68, no. 7, pp. 3268–3275, Jul. 2021.

[25] V. K. Verma and J. N. Tripathi, "Analytical modeling of deterministic jitter in CMOS inverters," *IEEE Trans. Signal Power Integr.*, vol. 2, pp. 64–73, 2023.

[26] J. N. Tripathi, R. Achar, and R. Malik, "Fast analysis of time interval error in current-mode drivers," *IEEE Trans. Very Large Scale Integration (VLSI) Syst.*, vol. 26, no. 2, pp. 367–377, Feb. 2018.

[27] H. Park et al., "Power supply induced jitter (PSIJ) modeling, analysis, and optimization of high bandwidth memory (HBM) I/O interface," in *Proc. IEEE Symp. Electromagn. Compat., Signal Power Integr.*, 2023, pp. 125–130.

[28] J. N. Tripathi, R. Achar, and R. Malik, "Efficient modeling of power supply induced jitter in voltage-mode drivers (EMPSIJ)," *IEEE Trans. Compon., Packag., Manuf. Technol.*, vol. 7, no. 10, pp. 1691–1701, Oct. 2017.

[29] J. N. Tripathi, P. Arora, H. Shrimali, and R. Achar, "Efficient jitter analysis for a chain of CMOS inverters," *IEEE Trans. Electromagn. Compat.*, vol. 62, no. 1, pp. 229–239, Feb. 2020.

[30] X. J. Wang and T. Kwasniewski, "Propagation delay-based expression of power supply-induced jitter sensitivity for CMOS buffer chain," *IEEE Trans. Electromagn. Compat.*, vol. 58, no. 2, pp. 627–630, Apr. 2016.

[31] S. S. Santhanam, A. Dhole, and S. Anumala, "Generic methodology and solution for additive jitter correction in mesh-based clock architectures," in *Proc. Int. Symp. VLSI Des. Test*, 2024, pp. 1–6.

[32] Y. Sun, M. Ouyang, X. Sun, and C. Hwang, "Prediction of power supply induced jitter with PDN design parameters," *IEEE Trans. Electromagn. Compat.*, vol. 64, no. 6, pp. 2238–2248, Dec. 2022.

[33] C. Hwang, J. Kim, B. Achkir, and J. Fan, "Analytical transfer functions relating power and ground voltage fluctuations to jitter at a single-ended full-swing buffer," *IEEE Trans. Compon., Packag., Manuf. Technol.*, vol. 3, no. 1, pp. 113–125, Jan. 2013.

[34] L. Wu, "Prediction of PSIJ in DDR interfaces considering higher-order effects," in *Proc. DesignCon*, 2024.

[35] R. Wolff, "Output buffer PSIJ analysis," in *IBIS ATM Task Group Meeting*, 2021, pp. 1–23.

[36] Z. Yang, S. Huq, V. Arumaghum, and B. Ross, "BIRD95.6: Power integrity analysis using IBIS," Oct. 2005. [Online]. Available: http://ibis.org/birds/bird95.6.txt

[37] Y. Sun and C. Hwang, "Improving power supply induced jitter simulation accuracy for IBIS model," in *Proc. IEEE Int. Symp. Electromagn. Compat.*, 2021, pp. 1127–1132.

[38] Y. Ding, Y. Sun, R. Wolff, Z. Yang, and C. Hwang, "IBIS model simulation accuracy improvement by including power-supply-induced jitter effect," *IEEE Trans. Signal Power Integr.*, vol. 3, pp. 21–29, 2024.

[39] Y. Ding, Y. Sun, R. Wolff, Z. Yang, and C. Hwang, "IBIS model simulation accuracy improvement by nonlinear power supply induced jitter effect," *IEEE Trans. Signal Power Integr.*, vol. 4, pp. 55–64, Mar. 2025.

[40] Y. Ding, C. Hwang, Z. Yang, A. Muranyi, and R. Wolff, "BIRD 220.2: Pre-driver PSIJ keyword," Jan. 2025. [Online]. Available: https://ibis.org/birds/

[41] X. K. Cai, C.-T. Chen, and E. J. Cheng, "IBIS-approved streamlined power integrity model (SPIM) for platform power integrity analysis," in *Proc. IEEE Int. Symp. Electromagn. Compat., Signal Power Integrity*, 2024, Art. no. 414.

[42] K. Cai, F. N. Tan, C.-T. Chen, and M. Mirmak, "BIRD 226: PSIJ sensitivity," Aug. 2023. [Online]. Available: https://ibis.org/birds/

[43] M. Chang, "A novel Laplace transform modeling for supply induced jitter sensitivity of SERDES transmitter," in *Proc. IEEE 5th Int. Conf. Integr. Circuits Microsystems*, 2020, pp. 168–172.

[44] Z. Xu, Z. Wang, Y. Sun, C. Hwang, H. Delingette, and J. Fan, "Jitter-aware economic PDN optimization with a genetic algorithm," *IEEE Trans. Microw. Theory Techn.*, vol. 69, no. 8, pp. 3715–3725, Aug. 2021.

[45] L. Jiang, L. Zhang, and E.-P. Li, "PDN noise-jitter co-optimization using physics-assisted genetic algorithm," in *Proc. IEEE Int. Symp. Electromagn. Compat., Signal Power Integrity*, 2024, pp. 526–531.

[46] Y. Sun, J. Kim, and C. Hwang, "Jitter-aware target impedance," in *Proc. IEEE Int. Symp. Electromagn. Compat., Signal Power Integrity*, 2019, pp. 217–222.

[47] Y. Sun, J. Kim, M. Ouyang, and C. Hwang, "Improved target impedance concept with jitter specification," *IEEE Trans. Electromagn. Compat.*, vol. 62, no. 4, pp. 1534–1545, Aug. 2020.

[48] J. Kim et al., "Improved target impedance and IC transient current measurement for power distribution network design," in *Proc. IEEE Int. Symp. Electromagn. Compat.*, 2010, pp. 445–450.

[49] J. Kim, T. Yuzo, K. Araki, and J. Fan, "Improved target impedance for power distribution network design with power traces based on rigorous transient analysis in a handheld device," *IEEE Trans. Compon., Packag., Manuf. Technol.*, vol. 3, no. 9, pp. 1554–1563, Sep. 2013.

[50] B. Mutnury, M. Swaminathan, M. Cases, N. Pham, D. N. d. Araujo, and E. Matoglu, "Modeling of power supply noise with non-linear drivers using recurrent neural network (RNN) models," in *Proc. Electron. Compon. Technol.*, 2005, pp. 740–745.

[51] B. Mutnury, M. Swaminathan, and J. P. Libous, "Macromodeling of nonlinear digital I/O drivers," *IEEE Trans. Adv. Packag.*, vol. 29, no. 1, pp. 102–113, Feb. 2006.

[52] M. Chang, "Machine learning-based verilog-a modeling for supply induced jitter sensitivity of high-speed memory interface: Two layer PCB case study," in *Proc. IEEE Int. Joint EMC/SI/PI EMC Europe Symp.*, 2021, pp. 44–47.

[53] A. Javaid, R. Achar, and J. N. Tripathi, "Development of knowledge-based artificial neural networks for analysis of PSIJ in CMOS inverter circuits," *IEEE Trans. Microw. Theory Techn.*, vol. 71, no. 4, pp. 1428–1438, Apr. 2023.

[54] A. Javaid, R. Achar, and J. N. Tripathi, "Prediction of power supply induced jitter via deep belief and knowledge-based neural networks," in *Proc. IEEE 28th Workshop Signal Power Integrity*, 2024, pp. 1–3.

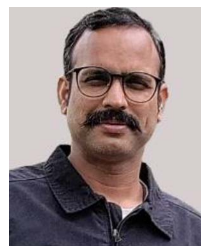
[55] T. Shin et al., "Modeling and analysis of system-level power supply noise induced jitter (PSIJ) for 4 Gbps high bandwidth memory (HBM) I/O interface," in *Proc. IEEE Elect. Des. Adv. Packag. Syst.*, 2021, pp. 1–3.

[56] C. Zhi et al., "Multiobjective optimization for PSIJ mitigation and impedance improvement based on PCPS/DR-NSDE in chiplet-based 2.5-D systems," *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.*, vol. 43, no. 8, pp. 2235–2248, Aug. 2024.

Chulsoon Hwang (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from the Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, in 2007, 2009, and 2012, respectively.

From 2012 to 2015, he was a Senior Engineer with Samsung Electronics, Suwon, South Korea. In July 2015, he joined the Missouri University of Science and Technology (formerly University of Missouri-Rolla), Rolla, MO, USA, where he is currently an Associate Professor. His research interests include RF desense, signal/power integrity in high-speed digital systems, EMI/EMC, hardware security, and machine learning.

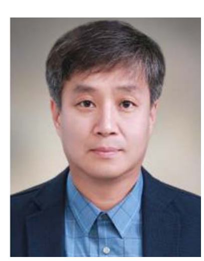
Dr. Hwang was a recipient of the AP-EMC Young Scientist Award, the Google Faculty Research Award, and Missouri S&T's Faculty Research Award, a corecipient of seven Best Paper/Best Student Paper Awards from various conferences including IEEE EMC+SIPI, AP-EMC, and DesignCon.



Jai Narayan Tripathi (Senior Member, IEEE) received the Ph.D. degree in electrical engineering from the Indian Institute of Technology Bombay, Mumbai, India, in 2014.

He was with STMicroelectronics, India, for almost 7 years. He was a Visiting Postdoctoral Fellow, in 2016, and a Visiting Scientist, in 2016 and 2017, with the Politecnico di Torino, Turin, Italy. From 2023–2024, he was a Visiting Professor with the Department of Electronics and Telecommunications, Politecnico Di Torino, Italy. He is currently an Associate Professor with the Indian Institute of Technology Jodhpur, Jodhpur, India, as well as an Adjunct Research Professor with Carleton University, Ottawa, ON, Canada. He has authored or coauthored more than 100 research papers in referred journals, as book chapters and in the proceedings of top international conferences. His current research interests include signal integrity, power integrity, electromagnetic interference/electromagnetic compatibility, metaheuristic optimization, and RF circuits.

Dr. Tripathi was the recipient of a couple of sponsored research grants as well as SIRE Fellowship from the Science and Engineering Research Board (SERB), Govt. of India, the Best Paper Awards in a couple of international conferences, the YITP Research Award by ACRI, Italy, in 2016 and 2017. He has served as a TPC Member for over 30 international conferences including the premier conferences such as IEEE EPEPS, IEEE SPI, IEEE VLSI Design, IEEE EDAPS, etc. He was an Invited Speaker in IEEE EDAPS 2015, Seoul, South Korea. He has served as a TPC CoChair for IEEE EDAPS 2018 and IEEE EDAPS 2024. He has been a Session Chair in several international conferences. He has been a referee for many IEEE journals. He was a member of the "Best Paper Award Committee" of IEEE EDAPS 2020. He has delivered invited talks at various universities and international forums. He is currently an Associate Editor for IEEE TRANSACTIONS ON SIGNAL AND POWER INTEGRITY, IEEE TRANSACTIONS ON COMPONENTS, PACKAGING, AND MANUFACTURING TECHNOLOGY, and IEEE ELECTROMAGNETIC COMPATIBILITY MAGAZINE. He was also an Associate Editor for the IEEE OPEN JOURNAL OF CIRCUITS AND SYSTEMS.



Dan Oh received the B.S., M.S., and Ph.D. degrees in electrical engineering from the University of Illinois Urbana-Champaign, Urbana-Champaign, IL, USA, in 1990, 1992, and 1995, respectively.

He worked in the fields of signal and power integrity at several leading technology companies, including Synopsys (1996), Rambus (2000), and Intel (2010). From 2016 to 2024, he led advanced packaging development initiatives at Samsung Electronics, including 2.5-D silicon/RDL interposers, fan-out wafer-level packaging (FO-WLP), fan-out panel-level packaging (FO-PLP), and 3-D through-silicon via (TSV) integration. He is currently an Engineering Director of Signal Integrity/Power Integrity (SI/PI), Product and Test Engineering, Celestial AI, Santa Clara, CA, USA. He holds over 150 patents and patent applications. He has authored more than 180 papers published in IEEE journals and conferences. He is also the lead author of the book *High-Speed Signaling: Jitter Modeling, Analysis, and Budgeting.*

Dr. Oh was a Corporate Vice President and later an Executive Vice President during his tenure with Samsung Electronics.